Probing the limit of forward materials screening and the novelty claimed by inverse design with 14.6 million crystals resolved across 55 databases

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Abstract

Materials discovery follows two routes. Forward screening searches known crystals for one with the property wanted; inverse design generates a crystal to order and reports how many of its products are new. Both are measured against public databases that cannot be measured against one another: the same crystal carries a different identifier in each, and the formula cannot serve as a key because polymorphs share one. We resolve identity at the level of the crystal structure across 55 public databases, keeping the source and the method behind every value. The record holds 14,590,485 distinct crystals and 131.8 million property values, and it grows as it ingests: a structure matching nothing becomes the reference the next source is matched against. What a forward screen inherits from the database it loads is then worse than that database can report. Independent sources computing one formation energy at one level of theory agree within 50 meV atom^{-1} only 65.3% of the time, as a systematic offset rather than noise; and of fifty properties needing a calculation of their own, none has been computed for a tenth of the pool. The same record is the reference a novelty claim needs. Auditing nine released sets of generated crystals against it, the databases holding the few matches are not the repository such claims are checked against, and the reported metric hides the behaviour it is meant to rank: a generator 71% unrecognised proposed a composition nobody had computed once in 500, where random enumeration, 89% unrecognised, did so 82 times in 492. A novelty fraction measures the reference and the criterion as much as the generator. This record supplies both, and 513 bounded learning tasks besides.

Keywords: materials databases, data integration, crystal structure deduplication, provenance, cross-database benchmarking, machine learning for materials

1 Introduction

Materials discovery follows two routes. Forward screening searches the crystals already known for one that has the property wanted. Inverse design generates a crystal to order, and reports as its main figure of merit how many of its products are new. Each route is usually described as bounded by something of its own: forward screening by how many crystals are known, inverse design by how good the generator is. Neither bound is where it is placed. Both routes rest on one question, asked of every crystal they touch: is this structure already known, and is this property already known for it? No public database can answer that question on its own.

The reason is that the databases share no key at the level of the crystal. Public repositories hold millions of computed structures [1–8], and generative models and universal interatomic potentials are trained on them [9–16]. But the same arrangement of atoms carries a different accession number in every database that holds it, and the chemical formula cannot serve as a key because polymorphs share one. So the union cannot be deduplicated, values from different providers cannot be placed on the same object, and three things stay unmeasured: how much of the record is one crystal counted several times, whether two sources that computed the same quantity agree, and which properties are known for which crystals. OPTIMADE gave the databases a common query interface [17, 18] and LeMaterial published a deduplicated union of several large ones [19]. Neither delivers the property layer, because two databases publishing a column called band gap are often not reporting the same physical quantity, and a union that merges by column name averages a semilocal, a hybrid and a measured value into one number.

That absence bounds each route in a specific way. A forward screen can rank crystals that nobody has computed, since that is what a surrogate model is for. What it cannot do is reach beyond the pool it was given or trust its labels more than the sources that produced them. A campaign screens whichever database it loaded and trains on that database's values, so the limit of forward screening is set by the size of the pool and the consistency of the labels, and both come from the record. Inverse design reports novelty against a reference assembled from a few of the largest repositories [19, 20], so the limit of what it may call new is set by the reference and by the criterion used to match against it, and both again come from the record. Three recent assessments make the point. A survey of one campaign's 384,870 stable predictions found them to be modifications of known materials [21]. The flagship compound of a generative model was later identified as a disordered phase reported in 1972 and present in the model's own training set [22]. And an ion-exchange

baseline built from known compounds outperformed four generative techniques [23]. In each case the record already contained what was reported as new.

Here we build the record that both routes need. Identity is resolved across 55 databases at the level of the crystal structure. Every value keeps the source, the native record and the method that produced it, so that several sources speaking about one crystal stay separately audible. Licence boundaries are enforced in the pipeline. Admission and recognition are one operation: a record that matches nothing becomes the corpus's next member, and its fingerprint joins the criterion applied to every record after it, so the corpus grows as it ingests rather than being rebuilt. What the record then shows is that neither route is bounded where it is usually placed. A forward screen inherits labels that disagree by more than the margin used to call a compound stable and a pool of which almost a quarter is invisible to any single provider, so its limit is set by the record and not by the size of the search. And a generative model's novelty fraction turns out to be a statement about the reference it was measured against: two published sample sets that the reported metric separates by eighteen points are separated by a factor of eighty once the question is whether anybody had computed the composition before.

2 Results

2.1 One crystal, one row, across 55 databases

Making 55 databases comparable comes down to one decision, taken once per incoming record: is this crystal already in the corpus? We settle it on the structure rather than on the label. Each record is matched first on a cross-database identifier the source is entitled to assert, then on a structural fingerprint, and never on the chemical formula, which polymorphs share (Fig. 1a; Methods). $CaMg_2Sb_2$, the crystal drawn there, ends the process as one row carrying 21 references from 17 of the 55 databases, no two of which name it the same way. A record that matches nothing is admitted rather than discarded, and every value it carries keeps the source, the native record and the method behind it. That second choice has a consequence this section measures: because an admitted record joins the criterion applied to the next, what results is not a merge of 55 databases but a corpus that extends the substrate it recognises them against.

The corpus so assembled holds 14,590,485 distinct crystal structures, of which 357,127 are experimental and 14,233,358 are computational candidates. On that pool sit 131,824,881 property

values spanning 166 properties, 1,198,897 stored spectra and 21,543,633 references recording which database contains which crystal.

The contributors are extremely unequal, spanning 5.7 orders of magnitude, with the largest supplying 46.0% of all values and 30 of the 55 supplying fewer than 10^5 each (Fig. 1b). The two ends of that distribution do different work: the large general-purpose databases determine how much of the pool carries any property at all, while the small specialised libraries determine how many distinct properties exist to carry [24–26] — the most property-diverse contributor supplies 61 of the 166 properties, the largest supplies 10.

The corpus is considerably larger than the pool it started from, and the difference is exactly the records that matched nothing. The deduplicated starting pool held 8,225,138 structures, and admission added 6,365,347 more, a growth of 77.4%.

What that union is made of differs completely from one database to the next, and the difference is invisible from inside any of them (Fig. 1c). Of the 5,894,222 crystals Alexandria annotates, 75.4% are annotated by no other database; of the 211,854 the Materials Project annotates, 1.3% are. The enumerated framework libraries sit at 100%, describing crystals nobody else has computed, while several specialised libraries sit below 1%, reporting only on crystals everybody already holds. A campaign that loads one database is therefore not working with a smaller copy of the record but with a differently shaped piece of it, and no query inside that database can reveal which.

Of the 21,543,633 references, 6,784,564 were resolved by record identifier, 8,294,560 by structural fingerprint alone, and the remainder record a database's own entry (Fig. 1d). The two matched routes are not interchangeable: identifier matching is exact but available only where a database publishes a cross-reference it owns, while fingerprint matching is available everywhere but has a domain of validity.

That domain has an edge, and it is worth stating before any conclusion is drawn from the corpus. Enumerated framework libraries are generated by decorating a small number of parent topologies, so sibling entries are identical in everything a bonding-graph fingerprint reads and more than 10^5 distinct hypothetical frameworks would collapse onto a few hundred pool materials. Such libraries are detected at the level of the source rather than the record and admitted under a composite key, and the boundary of validity this establishes is stated in Supplementary Note S2.

A structure admitted from one database becomes recognisable to every database that follows, because its fingerprint and identifiers join the matching substrate as it is admitted. The corpus therefore does not merely accumulate: the criterion by which it recognises the next record is itself a

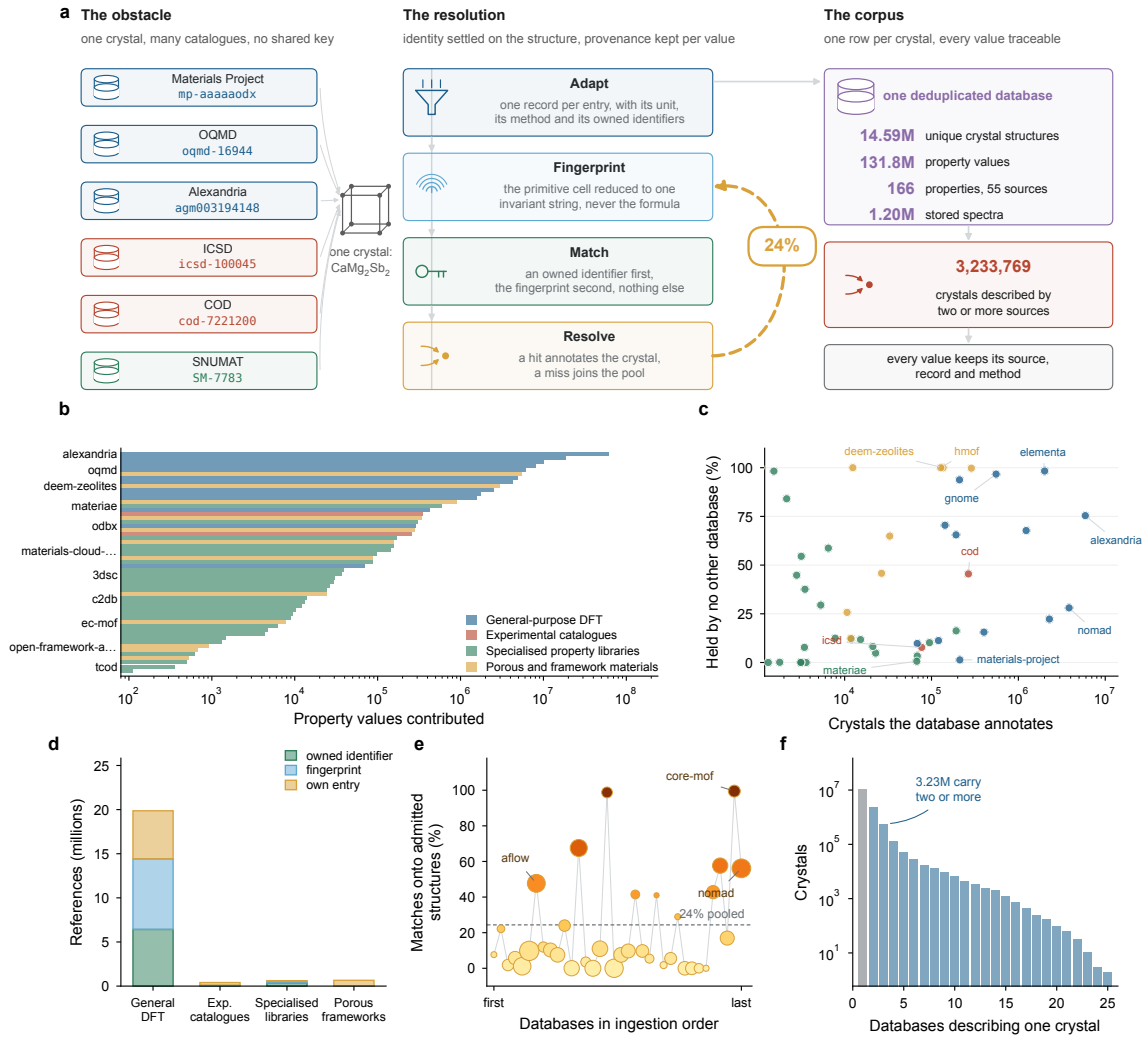


Fig. 1 Making the source databases comparable at the level of the crystal. a, The obstacle and the resolution. Each database labels the same arrangement of atoms with its own accession number, and no key relates them. Every incoming record is reduced to a structural fingerprint and matched first on a cross-database identifier the source is entitled to assert, second on that fingerprint, never on the chemical formula. The identifiers shown are the ones those databases publish for the crystal drawn, $CaMg_2Sb_2$. The dashed loop is the mechanism that makes the corpus self-evolving: a record matching nothing is admitted, and its fingerprint re-enters the matcher, so each source is matched against a larger pool than the one before it. The figure on the loop is the percentage of all matches that land on a crystal admitted this way. The right column reports what the assembled corpus holds. **b**, Property values contributed by each database on a logarithmic axis, coloured by family, with labels at regular intervals through the ranking. **c**, For each database, the crystals it annotates against the share of them that no other database annotates. Colour marks the database family. **d**, How the references were resolved, grouped by database family: by an identifier the source owns, by structural fingerprint, or as a record that the database contains the structure itself. **e**, For each database, the share of its matches that landed on a crystal admitted during assembly rather than present in the starting pool, in ingestion order. Marker area scales with the number of matches and the dashed line marks the pooled share. **f**, Number of databases describing one crystal. Grey marks the crystals held by a single database, blue those held by two or more.

product of the records already ingested, and it is by this mechanism that the database keeps itself current as sources publish new versions and new sources appear. The effect is measurable. Of the 15,079,124 references resolved by matching, 3,676,707, or 24.4%, land on a crystal that was absent from the starting pool and was admitted during assembly (Fig. 1e). Those links do not exist in a construction that matches every database against a fixed reference. The figure is a property of the assembly rather than of the sources, since a different ingestion order would distribute it differently between them, and it is a lower bound on what a frozen reference would have missed rather than a measure of quality. What it establishes is that such links exist at all, and how much of the record they reach.

The effect is ordered by ingestion time and by chemical distance from the starting pool. AFLOW, ingested shortly after Alexandria had contributed 1,514,488 structures, resolved 47.7% of its 1,401,895 matches onto crystals admitted during assembly; the enumerated framework libraries show it in its purest form, CoRE MOF resolving 99.5% of its matches onto frameworks four earlier libraries had placed in the pool; and databases whose chemistry the starting pool already covered sit at or near zero. The per-source figures are Supplementary Note S6. The practical consequence is that the marginal value of adding a database is not confined to the values it brings. CoRE MOF contributed 155,488 experimental porosity measurements, and the reason they landed on existing crystals rather than forming an isolated set of new ones is that earlier libraries had already put those frameworks in the pool. Assembled against a frozen reference the same measurements would have arrived as an unconnected island, and the corpus would have to be rebuilt from the beginning to connect them.

What the assembled record holds is therefore not a uniform gain but a long tail. Of the 13,902,836 structures carrying at least one reference, 3,233,769 carry two or more and a few hundred carry the account of twenty databases or more, while the single largest group is described by one database and no other (Fig. 1f). That distribution is what the next section measures against: it is both the only thing that makes a cross-database comparison possible and the reason a screen loading one database cannot see what it is missing.

2.2 What a forward screen inherits

A forward screen inherits two things from whichever database it loads: the labels it trains on and the pool it ranks. Neither can be measured from inside that database, because each is a property of the record as a whole, and measured on the resolved union both turn out to be worse than the

loaded database is able to report. Where two independent sources describe the same crystal they disagree by more than the margin conventionally used to decide whether a compound is stable, and almost a quarter of the labelled record is unavailable from any single provider. This section establishes the two, and they are the reason the completion of Section 2.3 has to be run over the whole record rather than over a slice of it.

The labels first. The redundancy just described is what makes a comparison possible at all, and it is concentrated: the largest single overlap between two databases holds 1,672,167 shared crystals (Fig. 2a). That redundancy and it also bounds what any cross-database evaluation can claim, since for the largest overlapping pairs a model trained on one repository and tested on another is tested on hundreds of thousands of crystals it has already seen unless the split is taken over structures. The distribution of that redundancy over databases and over degree is Supplementary Note S7.

Comparing the redundant copies is possible only because every value kept the method that produced it. Each method string is collapsed to a level of theory and two databases are held against one another only where they answer the same question, so that a pair meeting only across that boundary is not compared at all (Fig. 2b–d; Supplementary Note S7). On that basis 846,303 crystals carry a formation energy from two or more databases at one level of theory and 1,360,573 carry a band gap. Concordance falls as the quantity becomes harder to compute: within 50 meV atom^{-1}, 65.3% of formation energies and 64.2% of hull distances agree, within 50 meV 82.1% of band gaps do, and within 5 GPa 44.3% of bulk moduli do. The tail matters more than these medians. The tenth of shared formation energies that agree least differ by more than 205 meV atom^{-1} (Fig. 2c), several times the hull-distance margins conventionally used to call a compound stable [27, 28]. Whether a candidate passes such a filter can therefore depend on which database supplied its energy, and in a corpus without structural identity that dependence is not merely unquantified but unobservable.

Three features of the comparison sharpen that result rather than softening it, and each is established in Supplementary Note S7. Agreement between two databases is not evidence of independence: the pairs that agree most closely are those carrying one underlying calculation twice, so the concordant fractions above overstate the true level of corroboration. The disagreement is not a failure of the underlying theory, which independent implementations reproduce far more closely than this [29], but of the choices each provider makes around the solver, so it will not be removed by better software; for the major pairs it appears as an offset of up to 177 meV atom^{-1} carrying one sign, with correlations above 0.85. And it is not an artefact of comparing different quantities, since pairs

that meet only across a boundary of theory are excluded, which lowers the measured disagreement rather than raising it.

The pool is the second inheritance, and the question it raises is how much of it carries any label at all. The answer is: very little, and what there is was distributed neither by importance nor by cost. Of the 166 properties in the registry 151 carry at least one value, and treating the pool and those 151 as a matrix gives 2,203,163,235 cells of which 99,493,466 are occupied, so the matrix is 4.52% full (Fig. 2e). Occupancy is extremely uneven: nine properties cover more than 40% of the pool while 97 of the 151 cover less than 1%, and half of all crystals carry at most seven properties (Supplementary Fig. S1a). Only 66 of the 151 are reported by two or more databases; the other 85 have a single supplier and therefore no independent check of any kind.

The gradient follows price, and then departs from it in a way that identifies what it really follows (Fig. 2f). Of 69 general-purpose properties, the 19 that are read off a self-consistent calculation a database performs anyway have a median coverage of 4.56% of the pool, while the 50 that require a calculation of their own have a median of 0.107%, lower by a factor of 43, and *not one of the fifty reaches 10%*. Within that second group, however, the linear-response properties are the more expensive and reach a higher median coverage (1.06%) than those obtainable from a few extra self-consistent runs (0.029%), a thirty-six-fold inversion of what price predicts. The elastic and dielectric tensors were the target of dedicated high-throughput campaigns [30, 31]; quantities such as the work function, cheaper to obtain, were the target of none. What the coverage gradient records is therefore not the difficulty of a calculation but whether somebody once organised a campaign to perform it, and an expensive property that no campaign addressed is not evidence that the property is unimportant, only that its turn has not come. That map is legible only once the databases sit on one structure pool, because no single repository can report on effort it did not spend.

Sparse as the matrix is, part of it was filled by the integration itself, which is the section's second result (Fig. 2g). Every structure has exactly one contributor, the database that introduced it, and of the 6,266,517 structures admitted during assembly that carry any value, 1,637,460, or 26.1%, carry at least one value their own contributor never reported. The effect is largest where the chemistry is crowded, two thirds of the structures AFLOW contributed having gained a value it never published, and close to zero for the enumerated libraries of hypothetical frameworks, whose structures nobody else has computed. Because a database's own structures are by construction those no earlier source held, this figure depends on ingestion order; the order-independent statement is that 3,153,748 crystals, 22.9% of all that carry any value, draw their property record from more

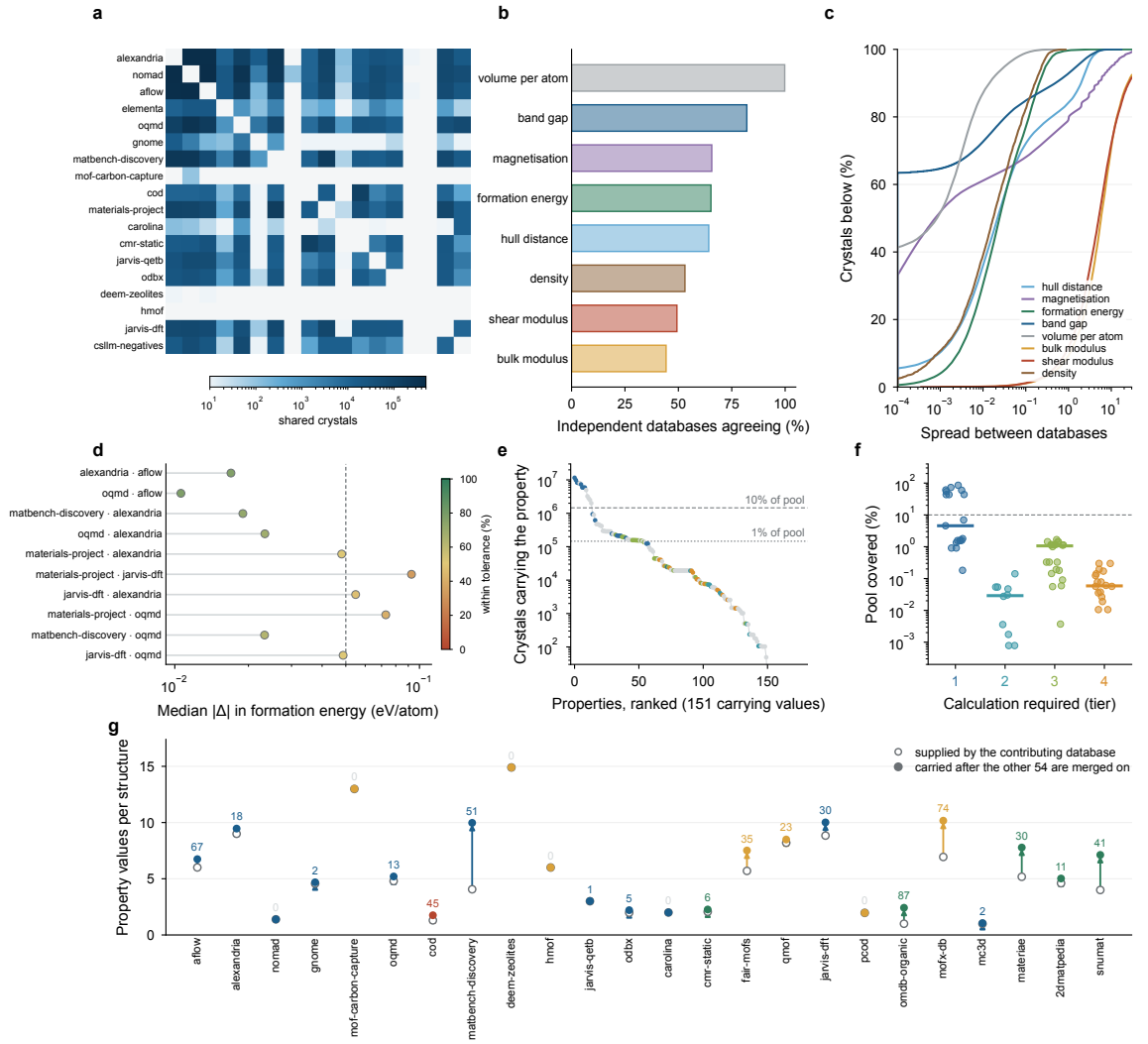


Fig. 2 What a forward screen inherits from the database it loads. a, Shared crystals between the databases annotating the most structures, on a logarithmic colour scale. An empty cell marks a pair with no crystal in common rather than missing data. **b**, For each comparable property, the fraction of multi-source crystals whose independent values fall within the tolerance stated in Supplementary Note S7. The figure beside each bar is the number of crystals carrying that property from two or more databases at one level of theory. **c**, Cumulative distribution of the spread between databases, in the property's own unit for energies, magnetisation and moduli and as a relative difference for density and volume per atom. **d**, Median absolute difference in formation energy for the database pairs sharing the most crystals, coloured by the fraction agreeing within the tolerance marked by the dashed line. **e**, Every property carrying at least one value, ranked by the number of crystals that carry it. Colour marks the calculation tier assigned in **b**. Grey marks properties outside that classification, whose domain is a single material family. The horizontal lines mark the stated fractions of the pool. **f**, Coverage of the general-purpose properties grouped by the calculation each requires: tier 1, readable from a single self-consistent run; tier 2, a few extra self-consistent runs such as a slab or a strained cell; tier 3, linear response or finite differences over many displacements; tier 4, anharmonic, transport or beyond-semilocal treatment. Each point is one property, horizontal bars are group medians and the figures above them give those medians. **g**, What each database's own structures gained from the others. Every structure has one contributing database, the one that introduced it; the open marker is the property record that database supplied and the filled marker what the structure carries once the others are merged onto it, averaged over the structures each contributed. The figure above each database is the percentage of its structures that gained at least one value it never reported. The databases contributing the most structures are shown.

than one database. Either way, almost a quarter of the labelled record is not available from any single provider, so a campaign that loads one database ranks a smaller pool than the one that exists and has no indication of what it is missing. The two inheritances bound a screen from opposite directions: the pool fixes what it can be asked about, and the labels fix how far its answer can be trusted.

2.3 Probing the limit of forward screening

Both bounds of Section 2.2 move together if the property layer is completed by prediction over the whole record rather than over one provider's slice of it. Whether such a completion succeeds is a question about trained models, and it is answered when their held-out numbers exist. What can be established now is that the problem is well posed on this corpus and is not well posed on any single database, and that is what this section does.

The model that consumes it is built for that record rather than for a benchmark (Fig. 3a). Each crystal enters as a periodic radius graph with lattice self-edges, and every interaction block does three things in turn: an attention over each edge taken in the frame of that edge, so that rotations are handled by construction rather than by augmentation; a tensor square of the node state, which lets one pass build the angular products a message-passing layer would otherwise need several passes to reach; and a reciprocal-space term, in which every atom emits a learned form factor and receives the origin-free Fourier sum of the others over a fixed set of Miller indices, coupling every pair in the cell and every periodic image at once rather than only the neighbours inside the cutoff. The crystal is read out twice, once as the diffraction intensities those form factors imply and once by attention over the atoms, and the head emits a value, an uncertainty and the domain the value is valid on. The reciprocal terms and the diffraction readout enter through zero-initialised gates, so the model at initialisation is the real-space model and the extra machinery has to earn its weight.

A completion needs three things no single database provides: structures deduplicated, so that a train–test split falls on distinct crystals rather than on records; labels that keep the quantity they denote rather than the column they were filed under; and, beside each label, the spread of the sources that reported it, which fixes the resolution at which a predicted value can be read, since a prediction closer to the truth than the sources are to each other cannot be shown to be so. The corpus supplies all three, and imposes four constraints on how they may be used. Filling the same gap by calculation would take an effort no single group can supply.

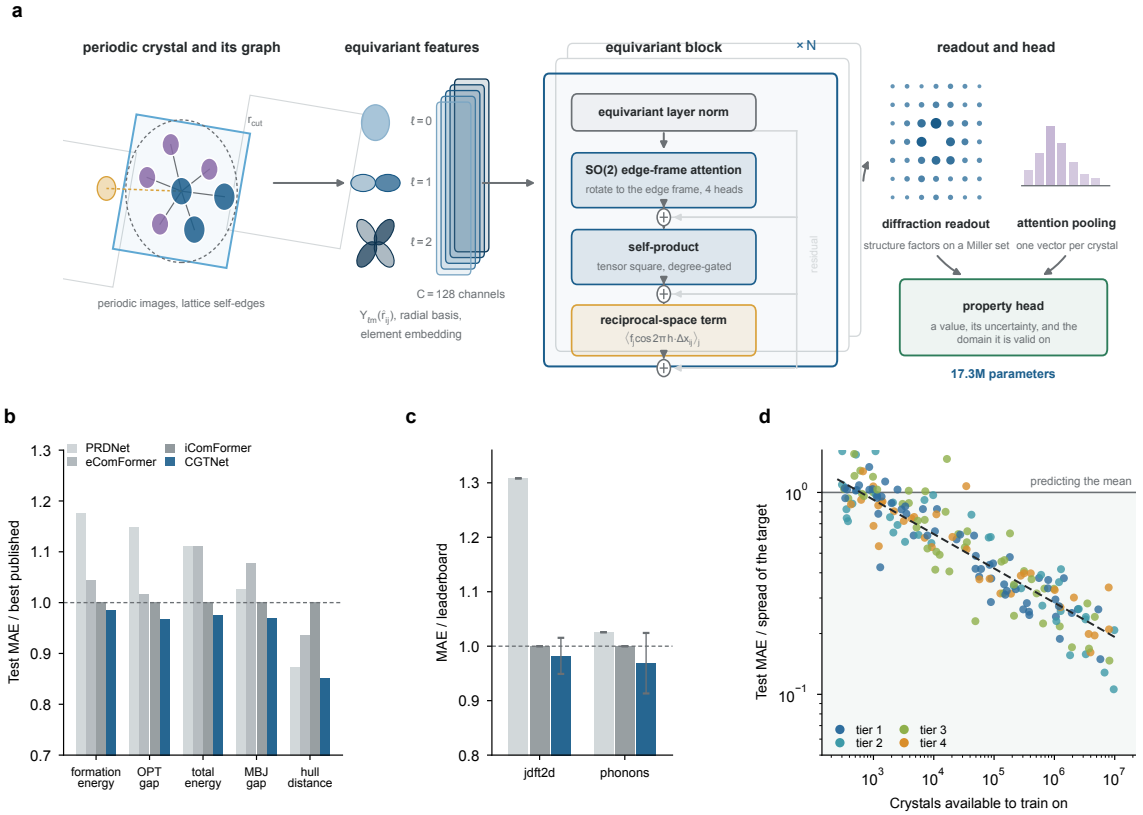


Fig. 3 The model trained on the record. a, One interaction block and the readout. A crystal enters as a periodic radius graph with lattice self-edges; the block applies an attention in the frame of each edge, a tensor square of the node state, and a reciprocal-space term in which each atom receives the origin-free Fourier sum of learned form factors over a fixed Miller set. The crystal is read out as diffraction intensities and by attention over atoms, and the head emits a value, an uncertainty and the domain the value is valid on. Grey rails are residual paths and the two lower terms enter through zero-initialised gates. **b**, Test error on the JARVIS-DFT tasks, as a ratio to the best published value for each task. **c**, Test error on the Matbench tasks, as a ratio to the leaderboard entry, with the spread over folds. **d**, Test error over every task of the corpus against the number of crystals available to train it, each error divided by the spread of its own target, coloured by the calculation tier of Fig. 2f. The line marks a model that predicts the mean.

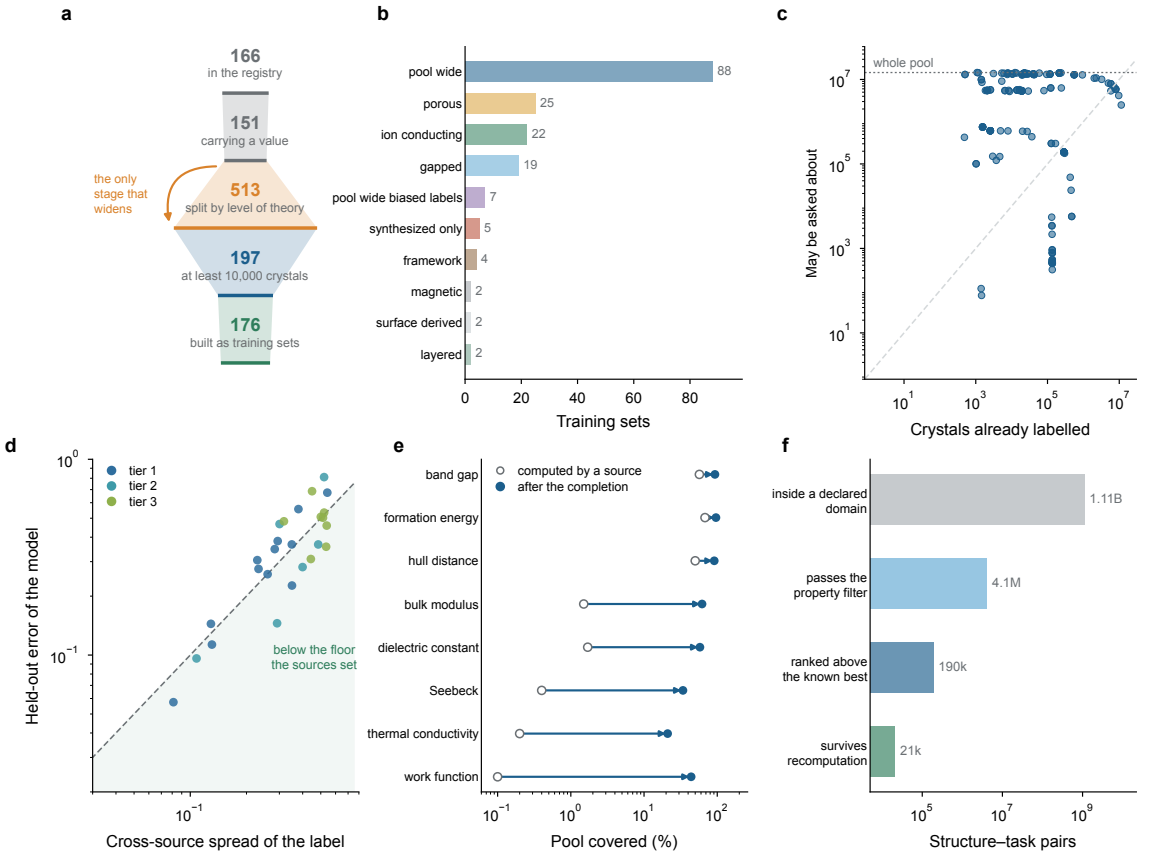


Fig. 4 What the corpus offers a learned completion of the property layer. a, How the registered properties become separable learning tasks and then realised training sets. The arrow marks the one stage that multiplies rather than filters, splitting a property name by level of theory. **b**, The constructed training sets by declared applicability domain, the set of structures for which the property is defined at all. **c**, For each training set, the crystals already labelled against the crystals the task may legitimately be asked about once its domain is enforced. The dotted line marks the whole pool and the dashed diagonal marks parity. **d**, Held-out error of each completed task against the spread between the sources that supplied its labels, coloured by calculation tier. The shaded region lies below the spread the sources themselves carry. **e**, Coverage of the pool for the most widely covered properties, as supplied by the sources and after the completion. **f**, What the completed layer nominates, and what survives each successive filter.

A census establishes the first half of that claim, which is that there is enough to train on (Fig. 4a). The unit that matters is not the property but the property at one level of theory, because a name such as band gap covers several different physical quantities that a model cannot distinguish from structure alone. Split that way, the 151 populated properties yield 513 candidate tasks, set against the tens that existing property-prediction and stability benchmarks provide [32, 33]. Of these, 165 regression tasks have at least 10^4 crystals to train on, 100 have at least 10^5, and 13 have at least 10^6. Constructing the training sets under a default policy that keeps the largest level of theory per property produced 176 sets covering 116 properties and 93,864,594 labels, each with a chemistry-aware train, validation and test split.

The four constraints are the second half, and each is a direct consequence of the integration reported above. The first is that a property name is not a label definition (Supplementary Fig. S2).

Band gap carries 15 distinct levels of theory, of which the semilocal group holds 83% and no other reaches 5%. The systematic offsets between them are of order 1 eV, which a model cannot infer from the structure and would instead learn as a proxy for which database supplied the row. The second is that some quantities are condition-dependent rather than multiply determined: carbon dioxide uptake is reported at several distinct grand-canonical conditions, so a material legitimately carries several different correct values and taking their median produces a number that is not any physical quantity. Neither the level of theory nor the condition is resolved away, and neither is the cross-source spread of Section 2.2: across the training sets, 5,856,861 crystals carry labels from two or more databases, and that spread is kept beside the resolved value so that it can weight the loss rather than being averaged into it.

The third is that the data cannot be split at random. The same compound recurs in the pool under different accession numbers, so a random split places identical chemistry in training and test, and the resulting scores are optimistic in exactly the regime that matters, since the crystals a completion is wanted for are the ones whose chemistry the training set never saw. Holding out whole chemical systems is the right default, but it degenerates on the enumerated libraries, where an entire task can occupy a single chemical system and a chemistry-level holdout leaves the validation and test folds empty. The split rule falls back for those tasks and records which level it used (Supplementary Note S9).

The fourth is that the completion has a domain (Fig. 4b,c). Carbon dioxide uptake is a grand-canonical adsorption quantity: it is not zero for a dense intermetallic, it is undefined, because the simulation cell has nowhere to insert a molecule. Of the 176 training sets, 63 draw more than 90% of their labels from a single material family (Fig. 4b). Enforcing their declared domains changes the size of the completion problem substantially: 109 of the 176 tasks are defined on less than half the pool, and the number of predictions that lie inside a declared domain is 1,105,593,143 rather than the 2.57 billion that asking every task about every crystal would produce, a factor of 2.3. The difference is not a loss of coverage. It is the removal of numbers that would have looked like data. Plotting what a task may be asked about against what it already knows separates the two situations that matter (Fig. 4c): a task lying far above the diagonal has the most to gain, because its domain is large and its labels are few, while a task lying to the right and low is one whose domain is already nearly labelled and where a model has little left to add.

One dependency in that structure is worth stating because it closes the loop with Section 2.1. Twenty-one of the tasks are gated on a property the database does not yet hold: whether a crystal

has a gap, or whether it is magnetically ordered, is unknown for 108,246,933 structure-task pairs, so their admissibility can only be decided once the corresponding tier-1 models have run. The completion therefore proceeds in stages, and each stage enlarges the domain over which the next one is defined, in the same way that each admitted structure enlarged the substrate against which the next database was matched.

What such a completion would settle is worth stating before it is attempted, because it is the reason the corpus was assembled rather than merely enlarged. Searching the known set for a material that already has the property wanted does not require that the property have been computed, since a surrogate model can rank crystals nobody has evaluated, and that is how such campaigns are run. What is bounded is the pool the ranking is taken over and the labels the ranking was learned from, and Section 2.2 measured both. Running the completion over the whole corpus, within the domains of Fig. 4b,c, moves them together: the pool becomes the deduplicated record rather than one provider's slice of it, and every prediction carries the spread of the sources that trained it.

That is what measures a quantity the field has not had a way to ask about: how much of the known record has been passed over for want of a calculation rather than for want of merit. The answer is consequential in either direction. A large yield would mean that forward screening has been operating far inside its own reach, with materials that can already be made waiting to be recognised. A small one would mean the reach is nearly exhausted, and that further progress has to come from designing structures rather than from finding them — which is the route the next section turns to.

[Pending: the trained models]

The models are still training and no number below is in hand, so nothing in this section is asserted from them. This block becomes the closing passage of Section 2.3 and reports Fig. 3b–d and Fig. 4d–f: the multi-task backbone over the tier-1 tasks and the fine-tuned heads over the smaller ones, accuracy against the published benchmarks and across every task of the corpus, held-out accuracy set against the multi-source noise floor of Section 2.2, the coverage the completed layer reaches for the properties of Fig. 2f, and the boundary within which a predicted entry may be used. It is written when the held-out numbers exist, so that no claim precedes its evidence.

[Pending: what the completion nominates]

The candidate list does not exist yet. When the models of Section 2.3 have run, this block reports, for the properties whose held-out error falls below the multi-source spread of Section 2.2, how many crystals already in the record are nominated for a property no source has ever reported for them, how many of the highest-ranked survive recomputation, and how many survive a search of the literature the databases do not index. Selecting the extreme of a ranked prediction selects on prediction error as well as on the property, so the ranking is taken on predicted value gated by ensemble spread, and the selection effect is reported rather than assumed away.

2.4 Judging the novelty claimed by inverse design

The record that bounds forward screening is also the reference inverse design needs, and the point of this section is that a reported novelty fraction measures that reference at least as much as it measures the model. A generative model can only be shown to have produced something new against a reference that is both large enough to contain what else has been computed and complete enough to include the databases nobody thinks to check; short of that, the fraction reported as new is a statement about the omission. How much that matters is easiest to see in two released sample sets matched in everything but the generator, both ternary throughout. Under the test the field reports, a language model trained on a repository snapshot and a random enumeration of compositions differ by eighteen points. Under the test of whether anybody had computed the composition before, they differ by a factor of eighty, and in the direction that says the language model never left the space it was shown. A generative model counts as new whatever is absent from the reference set it was given, so a structure missing from one repository but held by another is counted as new; the reference sets in current use are drawn from a handful of the largest repositories [19, 20], and the three assessments cited in the Introduction all turned on exactly that [21–23].

Two distinct failures sit behind those three assessments, and the corpus assembled here has already met both from the other direction. The first is a criterion too permissive, counting a known prototype under a new decoration as a new material. The second is a criterion that misses a match it should have made, which is what happens when an ordered approximant is generated for a parent phase that is disordered on one site. Section 2.1 reports the first as over-merging in the enumerated framework libraries, where a bonding-graph fingerprint collapses more than 10^5 distinct hypothetical frameworks onto a few hundred pool materials, and Supplementary Note S2 states the domain within which that fingerprint is valid. Both are failures of the criterion rather than of the generator, and no published novelty figure states which criterion produced it.

What this record offers the route is therefore not a better score but a reference that can be named, and a criterion that travels with the number. Two groups auditing the same generator against the resolved corpus, under the same matching rule and the same stated domain of validity, obtain the same novelty fraction; against a reference assembled from a few of the largest repositories under an unstated rule, they need not. That is the condition under which a novelty claim becomes a measurement rather than a report, and it is what the audit reported below is built to supply.

The audit fixes the reference and varies the criterion. Nine released structure sets were put through it (Fig. 5; Methods): the 554,056 crystals GNoME contributes to this corpus [11]; the 770 structures MatterGen publishes with its paper [10], together with 256 crystals generated here from its released unconditional checkpoint and 1,081 from the checkpoint conditioned on bulk modulus; and the 500-structure samples released under a common relaxation protocol for CDVAE [9], CrystaLLM [34], FTCP, ion exchange and random enumeration [23]. The reference is all 55 sources. Four criteria are applied in turn, from strict to permissive: the crystal itself, under the structural fingerprint of Section 2.1; the reduced composition together with the space group; the reduced composition alone; and the stoichiometry pattern together with the space group (Fig. 5a). The three MatterGen sets agree to within 1.8 points under every criterion, so the unconditional run stands for them below and in the figure.

At the strict end the sets are largely new, and widening the reference does not change that as much as one would expect. Against Materials Project alone every set is 98–100% new; against all 55 sources the figures are 96.7% for GNoME, 95.0% for CDVAE, 89.4% for random enumeration, 86.8% for FTCP, 82.0% for MatterGen, 72.0% for ion exchange and 70.8% for CrystaLLM (Fig. 5b). Where the few matches sit is more informative than how many there are. Pooled over the six sample sets, the databases holding the structures a strict test still recognises are NOMAD (323), Alexandria (321), AFLOW (276) and OQMD (165); Materials Project, the reference such claims are most often taken against, holds 19 (Fig. 5c). For GNoME the asymmetry is sharper: of its 18,234 exact matches, Alexandria alone accounts for 14,709 and Materials Project for one, which is what deduplicating against one repository and not against another looks like from the other side.

Loosening the criterion moves the number far more than widening the reference does, and it is tempting to conclude that a structure should count as new only if it clears all four tests, on which reading almost nothing does: 0.0–8.2% of the sample sets and 45.1% of GNoME (Fig. 5d). That conclusion should be resisted, because the four criteria are not of one kind. The first three are decidable identity tests: a hit establishes that the record holds this crystal, or a crystal of this

composition in this space group, or a crystal of this composition. The fourth is not. Sharing a stoichiometry pattern and a space group is a necessary condition for two crystals to be the same prototype under a different decoration, not a sufficient one, so a hit establishes only that prototype identity cannot be excluded cheaply. Requiring a structure to clear it would call a genuinely new prototype known because some unrelated compound happens to share its anonymous formula and space group, and among 14.59 million crystals over 230 space groups almost nothing escapes that. The fourth column is therefore an upper bound on prototype redundancy rather than a test a claim must pass; settling it needs a structural comparison the fingerprint does not perform, and Supplementary Note S2 states what that would involve.

The strictest criterion that can be both decided and defended is the third, and it is the one a claim should carry: the fraction of a set whose reduced composition appears nowhere in the record. That fraction is 91.1% for GNoME, 60.0% for CDVAE, 34.0% for MatterGen, 25.8% for ion exchange, 16.7% for random enumeration, 3.0% for FTCP and 0.2% for CrystaLLM. Removing GNoME from the reference, so that one generated set is not used to judge another, moves none of these by more than 0.4 points. GNoME's 91.1% is not a discovery rate: 87.4% of its released set carries four or five elements against 29.2% of the record as a whole, and in that part of composition space a composition being absent carries little information about whether the crystal is distinct from what is known.

Two of these figures should be read together, because the sets behind them are matched: the CrystaLLM and random-enumeration samples are ternary throughout. Under the fingerprint the two look like a matter of degree, 70.8% against 89.4% absent from the record. At the composition level they are two orders of magnitude apart: 499 of CrystaLLM's 500 structures have a composition the record already holds, against 410 of random enumeration's 492. A language model trained on a repository snapshot rearranges compositions it was shown; random enumeration leaves that space. Neither behaviour is visible in the number that is usually reported, and a leaderboard ordered by that number places the two within twenty points of each other.

What the audit supports is therefore not a ranking of generators but a reporting convention. A novelty fraction is a property of three things, and only one of them is the model: the structures, the reference they were checked against, and the criterion that decided a match. Both of the other two are cheap to state and neither is usually stated. The resolved record supplies a reference that can be named and held fixed; the ladder supplies criteria that can be named and applied in turn; and reporting the fraction under each, rather than one figure under an unstated rule, is what turns a novelty claim into a measurement that a second group can reproduce. The requirement this places

on the reference is the one this work was built to meet. It is not that the reference be a repository of record, but that it be large enough that the relevant prior computation is in it at all, and complete enough that the aggregating and specialised databases which turn out to hold most of the matches are not the ones left out.

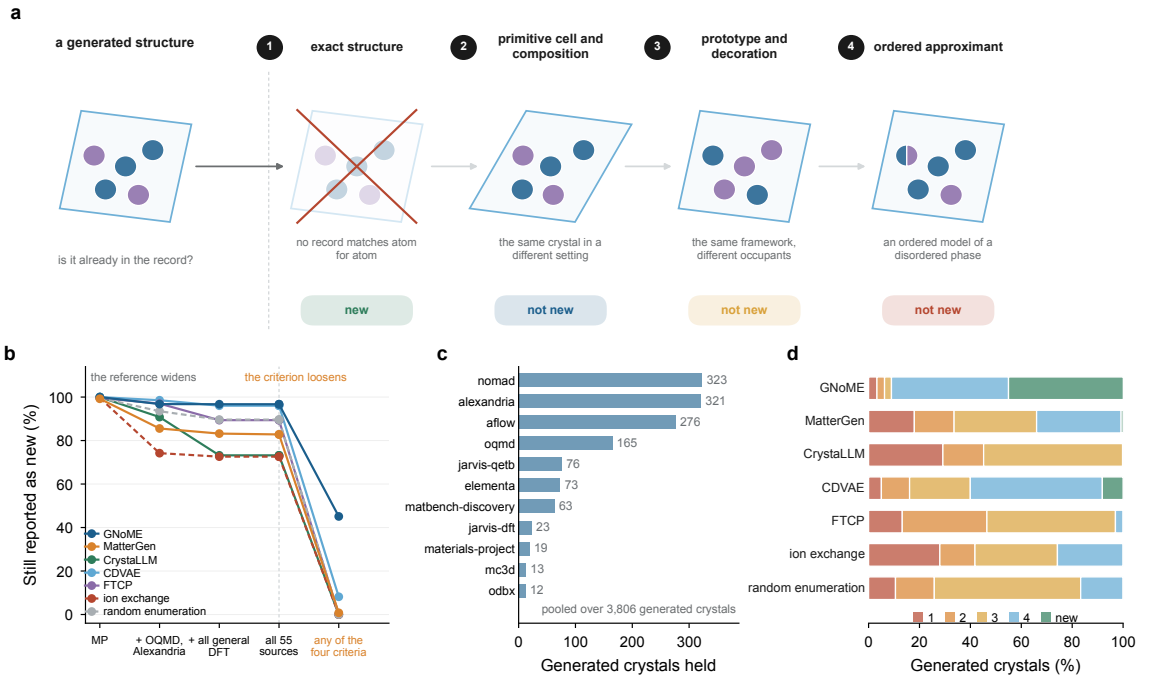


Fig. 5 What a claim of novelty is measured against. a, One generated crystal put to four identity criteria in turn, numbered as they are applied: whether the record holds this crystal, a crystal of this composition in this space group, any crystal of this composition, or any crystal sharing this stoichiometry pattern in this space group. Beneath each is the verdict that criterion returns for the crystal drawn. **b**, The share of each set that no match has yet been found for. Left of the rule the criterion is held at the first and the reference widens from one repository to every source; right of it the reference is held at every source and the criterion loosens to any of the four. Each position adds a test to the ones before it. **c**, The databases holding the crystals the first criterion recognises, pooled over the sample sets and the runs generated here. **d**, The strictest criterion under which each set is already in the record, as a share of the set. Colours are the criteria of **a**; *new* is the share that no criterion finds.

3 Discussion

Holding the public record as one structurally resolved space changes what can be asked of it. A campaign that used to begin with account requests, bulk downloads and format conversion begins with a query. A crystal is one row rather than six, so counting is possible, and every value names the calculation behind it, so comparison is possible. More consequentially, a repository can report only on its own contents, whereas a resolved union reports on the field, and two of the things it reports were not what a user of any single source would assume. Independent sources computing the same formation energy for the same crystal at the same level of theory agree within 50 meV atom^{-1} barely two times in three, and because the discrepancy is a systematic offset rather than scatter it does not average away with sample size: a corpus that merged the contributing columns would hand a model the offset to learn as though it were physics. In the other direction, sources that agree to the last decimal are usually carrying one calculation twice, so agreement between repositories should be read as shared provenance until shown otherwise. Neither observation requires new computation. Both require only that the record be held in a form where the same crystal is the same row and every number remembers where it came from.

For forward screening, the record moves the limit from where it is usually placed. The route is not confined to materials somebody has already computed, because a surrogate model ranks the uncalculated as readily as the calculated. It is confined by the pool it is asked about and by the labels it was taught from. A campaign screens one repository's holdings and trains on labels whose providers disagree by the margins reported above, and both of those are properties of the record rather than of the method. Running the completion over the whole corpus, inside the domains where each property is defined, moves both at once, and answers a question the field has not been able to pose: how much of the known record has been passed over for want of a calculation rather than for want of merit. Both answers are consequential. A large yield would mean forward screening has been operating far inside its reach, with already-synthesisable materials waiting to be recognised. A small one would mean the known set is close to exhausted and progress has to come from designing structures rather than finding them.

For inverse design, the record is the reference a claim of novelty needs, and the same record serves the route in two ways that are easy to conflate. It supplies what a generative model needs in order to work: deduplicated structures, and labels that name the quantity they denote, so that generation can be conditioned on a defined target rather than on a column heading. It also supplies what such a model needs in order to be believed. The failures reported against recent campaigns [21–23] were

not failures to generate but failures to recognise what the record already contained, and a claim of novelty is only as strong as the reference set and the identity criterion behind it. On this reading the two routes are not competitors to be ranked. Forward screening is bounded by how much of the known set it is asked about, inverse design by how much of the known set its novelty is measured against, and both bounds are the same missing thing seen twice. Supplying that record is what makes the two routes comparable, and eventually complementary, rather than separate literatures.

What makes this maintainable is that admission and matching are the same mechanism seen from two sides. A record that matches nothing is not an error to be logged but the corpus's next member, and once admitted it becomes part of the criterion by which the next record is judged. In this corpus 24.4% of all matches resolved onto crystals that assembly itself had added, approaching unity for chemistries the starting pool did not cover, and the last source ingested, matched against the largest substrate, found more than half of what it recognised among them. The same relation reappears in the completion stage, where twenty-one tasks can be gated only once earlier models have supplied the property their domain is defined by. A space built this way has no final state, which is what a merged snapshot cannot offer as its sources keep moving. Three boundaries should be read with the results. Structural fingerprinting has a domain of validity that enumerated framework libraries fall outside, and the composite key that rescues them is a remedy for a known failure rather than a general solution. Agreement is measurable only where two providers report the same quantity under comparable methods, so the disagreement reported here is measured on the part of the corpus most likely to be well behaved. And no value was recomputed: every number is as its source reported it, and the union inherits whatever errors those sources contain. What it adds is the ability to see when two of them disagree. The full set of limitations, including the licence boundary and the structural rather than geometric applicability gate, is Supplementary Note S10.

4 Methods

4.1 Sources and their ingestion

Each of the 55 databases is reached through one adapter emitting a stream of records, each carrying the source's native identifier, its property values with units and methods, its structure where the licence permits redistribution, and any cross-database identifiers the source is authoritative for. Sources that may not redistribute structures are marked properties-only in code, so no structure of theirs can enter the pool. The complete inventory, with each database's licence, contribution

and reference routes, is Supplementary Table S1. The adapter protocol and staging pipeline are Supplementary Note S1.

Property names are mapped onto a canonical registry of 166 entries, each with a fixed unit and value kind. A source field enters the registry under an existing name only when the quantity is the same physical quantity, and otherwise gains its own name. Method strings are required to distinguish values that a source reports for the same material under different settings, and the loader treats (native identifier, property, method) as the uniqueness key. The registry and its boundary cases are Supplementary Note S3.

4.2 Identity resolution

Structures are fingerprinted with the BAWL hasher [19] under the protocol used for the original deduplicated merge, so that fingerprints computed here are comparable with those already in the pool: parse, reject cells above 500 sites and disordered occupancies, reduce to the primitive cell with pymatgen [35], reject primitive cells above 300 sites, then hash.

Matching proceeds in two ordered paths against a map holding fingerprint-to-material and (source, identifier)-to-material entries. A record is looked up first by every authoritative cross-database identifier it carries, and then by fingerprint. Chemical formula is never a match criterion. A record matching nothing is admitted as a new pool material, and its fingerprint and identifiers enter the map, which is the mechanism measured in Section 2.1. Enumerated framework libraries, whose siblings are indistinguishable to a bonding-graph fingerprint, are detected at the level of the source and admitted under a composite key adding the cell volume per site (Supplementary Note S2).

Every match records the material, source, native identifier and route taken. Values are never overwritten or averaged: several databases reporting one property for one material coexist as separate rows.

4.3 Audit

Completeness is verified by three checks applied to each source independently: the row count the loader reported must equal the row count the database holds, every staged row must be either loaded or accounted for as an identifier no pool material corresponds to, and every property name present in the staged data must be present in the database for that source. The third catches a field the adapter maps but the loader never lands, which leaves row totals plausible while an entire quantity goes missing. The completed database passes all three for all 55 sources (Supplementary Note S4).

Every aggregate statistic reported here is computed with a single grouped scan over the property table. The natural alternative, a correlated subquery per source, rescans 132 million rows 55 times and evaluates each source against a different snapshot.

4.4 Cross-database comparison

Agreement was computed for eight properties whose values are comparable between databases once the level of theory is fixed, whereas extensive and convention-dependent quantities were excluded, because two databases disagreeing on a total energy reflects a different reference state rather than a different answer. Each method string is collapsed to a theory level and then to one of four comparable sets, and values are compared only within a set, so that a source holding several levels for one crystal is not measured against its own mixture and a crystal whose only overlap crosses a set boundary is not compared at all. Within each material and set, each source was reduced to the median of its own values, so a source reporting several method strings of one level is counted once, and the spread was taken as the range across sources. Tolerances, the excluded quantities and the full pair table are Supplementary Note S7 and Table S4.

The 69 general-purpose properties of Fig. 2f were assigned by hand to four tiers by the calculation each requires, excluding properties whose domain is one material family. The assignment is listed in full in Supplementary Table S3 so that it can be disputed item by item.

4.5 Census and training-set construction

A task is defined as one property at one level of theory, since method strings collapsed onto a single name denote different physical quantities. Under that definition the populated properties yield 513 candidate tasks, and a default policy keeping the largest level of theory per property realises 176 of them as training sets over 93,864,594 labels. Condition-dependent properties are split by condition rather than reduced to a median, cross-database spread is retained beside the resolved label, and splits are taken over chemical systems, falling back through formula to random where a task occupies too few systems to yield non-empty folds. Each task carries a declared applicability domain and an empirical one measured from its training set, and predictions are admitted only inside both. The full protocol, the four constraints it addresses and the domain assignments are Supplementary Note S9.

4.6 Novelty audit of generated structures

Released structure sets are read as deposited and fingerprinted under the protocol of the identity resolution above, so that a generated crystal is tested against the record by the same rule the record is deduplicated with. Structures that fail to parse, that exceed the site cap or that are partially occupied cannot be fingerprinted and are excluded from the set they belong to; the fractions reported are over the structures that could be. A set is then put to four criteria in turn, and each structure is filed under the strictest one that finds it: the fingerprint; the reduced composition together with the space group read from the fingerprint; the reduced composition alone; and the stoichiometry pattern together with the space group. The first three are identity tests on the crystal or on the compound. The fourth is a necessary condition for two crystals to be the same prototype under a different decoration and screens for prototype redundancy without deciding it, so it enters the audit as an upper bound; Supplementary Note S2 states what deciding it would require. Every match records which databases hold the matched crystal, so the same audit yields the fraction still unmatched against any subset of the sources. GNoME is itself one of the audited sets and also one of the 55 sources, so its own exclusive structures are withheld from the reference when it is audited, and each other set is reported both against the whole record and against the record with GNoME removed. MatterGen structures generated for this work were drawn from the released checkpoints under the sampling defaults, with the seed, batch size and checkpoint digest recorded beside the output.

4.7 Software and reproducibility

The pipeline is implemented in Python. Structure handling uses pymatgen [35] and symmetry analysis uses spglib [36]. The database is one SQLite file with two side-car binary blobs holding structures and spectra. Every figure is produced by a script reading comma-separated data files, and each of those files is produced by a named extraction script reading the published database read-only. No number quoted in this work was transcribed by hand.

Acknowledgements. We thank the maintainers of the 55 source databases integrated here. This work redistributes none of their data beyond what their licences permit, and depends entirely on their decision to publish in the first place.

Declarations

Competing interests. The authors declare no competing interests.

Data availability. The unified database is published as a versioned dataset on the Hugging Face Hub together with a development slice sized for interactive work. Five of the 55 sources are withheld from it: AFLOW's terms are academic and non-commercial, ELEMENTA and GNoME are CC BY-NC 4.0, Inorganic Crystal Structure Database records may be held only by a subscriber to the FIZ Karlsruhe licence, and NOMAD holds author-deposited calculations whose licence is set per upload rather than for the repository, so its values are withheld until the terms of each can be resolved. All five are searchable in a database built locally from the published pipeline and are filtered out before any public release. The database contains no ICSD structure. The comma-separated data files behind every figure are provided as source data.

Code availability. The acquisition adapters, the fingerprinting and matching pipeline, the loader, the completeness audit, the extraction scripts and the figure scripts are available in the repository accompanying this work.

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